Ketone

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Identity Sheet

Property	Value
Name	Ketone bodies
Other names	Ketones, acetone, acetoacetate (AcAc), beta-hydroxybutyrate (BHB)
Chemical formulas	Acetoacetate: $C_4H_5O_3^-$ (MW 101.08); BHB: $C_4H_7O_3^-$ (MW 103.10); Acetone: C_3H_6O (MW 58.08)
CAS numbers	AcAc: 541-50-4; BHB: 300-85-6; Acetone: 67-64-1
Appearance	Colourless in solution; acetone is volatile with fruity odour

Structural formulas:

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Acetoacetate: CH3-CO-CH2-COO(-)
Beta-hydroxybutyrate: CH3-CHOH-CH2-COO(-)
Acetone: CH3-CO-CH3
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Ketone bodies are produced by hepatic fatty acid beta-oxidation when carbohydrate availability is limited. The three ketone bodies are interconvertible: AcAc is the central species, reduced to BHB (by BHB dehydrogenase, NADH-dependent) or spontaneously decarboxylated to acetone. BHB is the predominant species in blood (~78%), AcAc (~20%), acetone (~2%). The dipstick primarily detects **acetoacetate**, not BHB.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Lactate	$C_3H_5O_3^-$	89.07	Product of anaerobic glycolysis, not ketogenesis
Pyruvate	$C_3H_3O_3^-$	87.06	Glycolysis intermediate
Free fatty acids	Variable	Variable	Precursors of ketones; not measured by ketone assays

Medical Information

Origin

Endogenous: Ketogenesis Ketone bodies are synthesised in **hepatocyte mitochondria** from acetyl-CoA derived from fatty acid beta-oxidation. Pathway: Acetyl-CoA -> acetoacetyl-CoA -> HMG-CoA (HMG-CoA synthase, rate-limiting) -> acetoacetate (HMG-CoA lyase). AcAc is reduced to BHB or decarboxylated to acetone. Ketogenesis increases during fasting, starvation, low-carb diets, uncontrolled diabetes, and prolonged exercise.

Exogenous Exogenous ketone supplements (BHB salts, ketone esters) are available as dietary supplements for ketogenic diet support and athletic performance. Ketogenic diets (very low carbohydrate) also promote ketogenesis.

Primary & Secondary Biological Roles

Primary role: - **Alternative fuel:** BHB and AcAc are oxidised by extrahepatic tissues (brain, heart, skeletal muscle) during glucose scarcity. Brain can derive up to 75% of energy from ketones during prolonged fasting.

Secondary roles: - **Signalling molecules:** BHB is a histone deacetylase (HDAC) inhibitor; anti-inflammatory signalling via HCAR2 receptor. - **Acetone:** Volatile waste product; exhaled via lungs (fruity breath odour).

Catabolism and Elimination Pathway

- BHB and AcAc are oxidised in peripheral tissues: BHB -> AcAc (BHB dehydrogenase) -> acetoacetyl-CoA (succinyl-CoA transferase) -> 2 acetyl-CoA -> TCA cycle.
- Acetone is exhaled via lungs or metabolised by CYP2E1.
- **Renal excretion:** Ketones are filtered and partially reabsorbed. At high plasma levels, reabsorption is saturated -> ketonuria.
- Normal urinary ketones: <5 mg/dL.

Expression in Humans

Normal Levels

Compartment	Reference Range
Blood BHB	<0.6 mmol/L (fasting up to 0.5 mmol/L)
Blood BHB (ketosis)	0.5-3.0 mmol/L (nutritional ketosis)
Blood BHB (DKA)	>3.0 mmol/L (often >5 mmol/L)
Urine ketones (dipstick)	Negative
Urine AcAc	<5 mg/dL

Factors Influencing Levels **Ketonuria (positive):** - Diabetic ketoacidosis (DKA) - Starvation / prolonged fasting - Alcoholic ketoacidosis - Ketogenic diet - Prolonged vomiting (starvation + dehydration) - Intense exercise - Pregnancy (accelerated starvation) - Glycogen storage diseases

Absent ketonuria: - Fed state with adequate carbohydrate - Well-controlled diabetes - Early DKA (BHB predominates but dipstick detects AcAc)

Associated Pathologies

Condition	Ketone Pattern	Key Symptoms
DKA	Strongly positive urine + blood BHB >3 mmol/L	Hyperglycaemia, acidosis, dehydration, Kussmaul breathing; ~5% of T1DM/year
Starvation ketosis	Mild-moderate ketonuria	Nausea, headache, fatigue
Alcoholic ketoacidosis	Moderate ketonuria + metabolic acidosis	Abdominal pain, vomiting after binge
Hyperemesis gravidarum	Ketonuria from starvation	Severe nausea/vomiting in pregnancy

Presence in Urine

Should they be normally present? No in fed state. Trace amounts may appear after overnight fasting or intense exercise.

Normal urinary levels: Negative (<5 mg/dL AcAc).

Form in urine: AcAc and BHB as their **anionic forms** at urine pH. Acetone is volatile and may evaporate.

Pathological significance:

Urinary Ketones	Possible Causes	Prevalence
Trace-small	Fasting, exercise, morning specimen	Common (benign)
Moderate-large	DKA, starvation, alcoholic ketoacidosis	DKA: ~5% T1DM/year

Solubility: All three ketone bodies are freely soluble in water. No crystallisation.

Detection in Urine

Available Clinical Assays

1. **Dipstick nitroprusside (Legal) test:**
   - **Principle:** Sodium nitroprusside (nitroferricyanide) reacts with AcAc in alkaline conditions -> purple colour. Does NOT detect BHB.
   - **Detection:** Colour change from beige to purple.
   - **LOD:** ~5-10 mg/dL AcAc (~0.5-1 mmol/L).
   - **Advantages:** Rapid, inexpensive.
   - **Disadvantages:** Does not detect BHB (dominant species in DKA); underestimates true ketosis.
2. **Blood BHB meter (point-of-care, preferred for DKA):**
   - **Principle:** BHB dehydrogenase on electrochemical strip converts BHB -> AcAc, generating electrons -> current.
   - **Detection:** Amperometric at BHB dehydrogenase electrode.
   - **LOD:** ~0.1 mmol/L.
   - **Advantages:** Detects BHB (dominant species); quantitative; preferred for DKA monitoring.
3. **Enzymatic BHB assay (laboratory):**
   - **Principle:** BHB + NAD+ (BHB dehydrogenase) -> AcAc + NADH. NADH measured at 340 nm.
   - **Detection:** 340 nm.
   - **LOD:** ~0.05 mmol/L.

Optimal Urine Type for Measurement

- **First morning urine** or random specimen for screening.
- For DKA monitoring, **blood BHB** is superior to urine ketones.
- Fresh sample: acetone evaporates and AcAc decomposes at room temperature.
- Cap specimen tightly; test within 1 hour.

Actual Gold Standard

Blood BHB measurement (enzymatic at 340 nm) is the gold standard for ketosis assessment. For urine, the **nitroprusside dipstick** remains the standard screening test, with the limitation of not detecting BHB.

Interferences in Measurement

Interference	Effect	Mechanism
Captopril, mesna	False positive (dipstick)	Thiol groups react with nitroprusside
L-DOPA metabolites	False positive	Colour interference
High specific gravity	False positive	Concentrated specimen
Ascorbic acid (high)	False negative	Inhibits colour reaction
Delayed testing	False negative	AcAc decomposes; acetone evaporates
BHB predominance	False negative (dipstick)	Dipstick detects AcAc, not BHB
Phenylketones (PKU)	False positive	React with nitroprusside

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis:** AcAc absorbs weakly at ~270 nm. Not useful in urine matrix.
- **Enzymatic NADH (340 nm):** See clinical assay. LOD: ~0.05 mmol/L.
- **NIR:** BHB and AcAc have C-H and C=O overtone bands (~1690-1730 nm region for C=O overtone). PLS calibration with NIR can estimate total ketones. LOD: ~0.5 mmol/L.

Fluorescence Detection

- **NADH fluorescence (enzymatic):** BHB dehydrogenase + NAD+ -> NADH (Ex 340 / Em 460 nm). LOD: ~0.01 mmol/L.
- **Amplex Red (H2O2 detection):** AcAc oxidised enzymatically to generate H2O2 -> resorufin (Ex 571 / Em 585 nm). LOD: ~0.005 mmol/L.

Raman Detection

Peak (cm-1)	Assignment
~1710	C=O stretch (AcAc, acetone)
~1365	CH3 symmetric deformation
~790	C-C stretch (acetone)
~1060	C-O stretch (BHB)

- LOD: ~10 mM (conventional Raman). SERS can improve to ~0.1 mM.
- Excitation: 785 nm.

FTIR Detection

Band (cm-1)	Assignment	Notes
~1720	C=O stretch (AcAc, acetone)	Distinguishable from urea C=O
~1240	C-O stretch (BHB)	–
~1370	CH3 deformation	–

- ATR-FTIR: LOD ~1-5 mM. Can detect ketonuria in DKA (levels often >5 mM).

Voltammetry Detection

- **BHB dehydrogenase biosensor:** Amperometric detection of NADH production at +0.3 V (mediator: Meldola Blue, prussian blue). LOD: ~10 uM.
- **Non-enzymatic AcAc reduction:** At Cu or Ni electrodes in alkaline media. LOD: ~50 uM.
- Linear range: 0.01-10 mM.

Other Detection Technologies

1. **Breath acetone analysers:** Gas sensors (semiconductor, SIER, photoionisation) detect exhaled acetone. LOD: ~0.1 ppm (correlates with blood BHB). Non-invasive.
2. **Paper-based colorimetric BHB:** BHB-DH + NAD+ + tetrazolium salt (colour at 530 nm). LOD: ~0.5 mM. Low cost.
3. **Wearable sweat ketone sensors:** Electrochemical BHB detection in sweat. Prototype stage.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
Dipstick (nitroprusside)	~5 mg/dL AcAc	Purple colour	None	Rapid, cheap
Blood BHB meter	~0.1 mmol/L	Amperometric	Fingerstick	BHB-specific, POC
Enzymatic BHB (lab)	~0.05 mmol/L	340 nm (NADH)	None	Quantitative
NADH fluorescence	~0.01 mmol/L	Ex 340/Em 460 nm	Enzyme	Very sensitive
FTIR	~1-5 mM	1720 cm-1	None	Reagent-free
Raman	~10 mM	1710 cm-1	None	Non-destructive
Breath acetone	~0.1 ppm	Gas sensor	None	Non-invasive
Voltammetry (BHB-DH)	~10 uM	+0.3 V mediator	Enzyme	Continuous

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References

1. RCPA Manual - Urine Dipstick. https://www.rcpa.edu.au/Manuals/RCPA-Manual/Pathology-Tests/U/Urine-dipstick
2. Wikipedia - Urine Test Strip. https://en.wikipedia.org/wiki/Urine_test_strip
3. StatPearls - Diabetic Ketoacidosis. https://www.ncbi.nlm.nih.gov/books/NBK534848/
4. StatPearls - Ketone Body Metabolism. https://www.ncbi.nlm.nih.gov/books/NBK493179/
5. PubChem - Acetoacetic acid, CID 96. https://pubchem.ncbi.nlm.nih.gov/compound/96